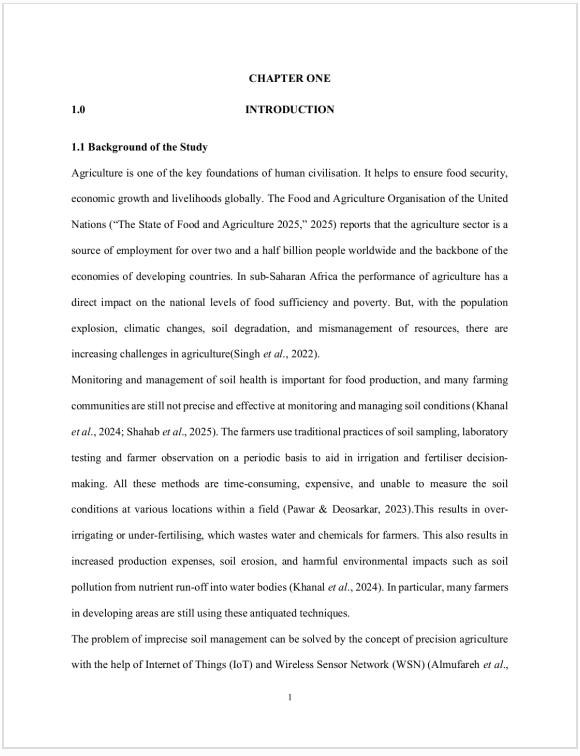
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CHAPTER ONE

1.0 INTRODUCTION

1.1 Background of the Study

Agriculture is one of the key foundations of human civilisation. It helps to ensure food security, economic growth and livelihoods globally. The Food and Agriculture Organisation of the United Nations ("The State of Food and Agriculture 2025," 2025) reports that the agriculture sector is a source of employment for over two and a half billion people worldwide and the backbone of the economies of developing countries. In sub-Saharan Africa the performance of agriculture has a direct impact on the national levels of food sufficiency and poverty. But, with the population explosion, climatic changes, soil degradation, and mismanagement of resources, there are increasing challenges in agriculture(Singh *et al.*, 2022).

Monitoring and management of soil health is important for food production, and many farming communities are still not precise and effective at monitoring and managing soil conditions (Khanal *et al.*, 2024; Shahab *et al.*, 2025). The farmers use traditional practices of soil sampling, laboratory testing and farmer observation on a periodic basis to aid in irrigation and fertiliser decision-making. All these methods are time-consuming, expensive, and unable to measure the soil conditions at various locations within a field (Pawar & Deosarkar, 2023).This results in over-irrigating or under-fertilising, which wastes water and chemicals for farmers. This also results in increased production expenses, soil erosion, and harmful environmental impacts such as soil pollution from nutrient run-off into water bodies (Khanal *et al.*, 2024). In particular, many farmers in developing areas are still using these antiquated techniques.

The problem of imprecise soil management can be solved by the concept of precision agriculture with the help of Internet of Things (IoT) and Wireless Sensor Network (WSN) (Almufareh *et al.*,

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